TT-S-178B
AMENDMENT-1
June 23, 1977

FEDERAL SPECIFICATION

SEALER, SURFACE RENEWER FOR WOOD AND SEALER FOR CONCRETE

This amendment, which forms a part of Federal Specification TT-S-178B, dated July 27, 1972, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

PAGE 1

Under Federal Specifications:

Change "TT-P-143" to "PPP-P-1892".

Add Paragraph 2.2 as follows:

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 56 - Test for Flash Point by Tag Closed Tester.
D 1308 - Test for Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 1544 - Test for Color of Transparent Liquids (Gardner Color Scale).
D 1545 - Test for Viscosity of Transparent Liquids by Bubble Time Method.
D 2369 - Test for Volatile Content of Paints.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

PAGE 2

Table I. Add to table I - under characteristics, "Lead, percent nonvolatile"; under minimum "---"; under maximum "0.06".

PAGE 3

Paragraph 4.3 and table II. Delete and substitute:

4.3 Test procedures. The sealer shall be tested as indicated in table II, and as specified hereinafter.

TABLE II. Index

Characteristics	Requirement Reference	Applicable Test: Fed. Test Method Std. No. 141	Applicable Test: ASTM Method	Paragraph Reference
Condition in container	3.2.1	---	---	4.3.1
Appearance	3.2.2	---	---	4.3.2
Storage stability, partially full container	3.2.3.1	---	---	4.3.3.1
Storage stability, full container	3.2.3.2	---	---	4.3.3.2
Odor	3.2.4	---	---	4.3.4
Resistance to cleaning solution	3.2.7.1	---	---	4.3.7.1
Resistance to hydrocarbon	3.2.7.2	---	---	4.3.7.2
Resistance to water	3.2.7.3	---	---	4.3.7.3
Resistance to wax solution	3.2.7.4	---	---	4.3.7.4
Nonvolatile matter	Table I	---	D 2369	---

FSC 8010

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TABLE II. Index

Characteristics	Requirement Reference	Applicable Test: Fed. Test Method Std. No. 141	Applicable Test: ASTM Method	Paragraph Reference
Color	Table I	----	D 1544	-----
Viscosity	Table I	----	D 1545	-----
Flash point	Table I	----	D 56	-----
Drying time	Table I	4061	----	-----
Abrasion resistance	Table I	6192	----	-----
Slipperiness	Table I	----	----	4.3.8
Lead content	Table I	----	----	4.3.11

Paragraphs 4.3.7.1, 4.3.7.2, 4.3.7.3, and 4.3.7.4. Delete "Method 6011 of Fed. Test Method Std. No. 141" and substitute "ASTM Method D 1308, Sec 5D".

Add new paragraphs:

4.3.11 Lead content.

4.3.11.1 Sample preparation. Using a 0.006-inch film applicator and a mechanical applicator plate, duplicate drawdowns for each sample of well-mixed paint shall be made on a standard paint penetration chart and dried for 24 hours. The drawdown shall be at least 10 inches long on the sealed portion of the penetration chart. The drawdown shall be cut into discs of appropriate size to fit the sample holder of a fluorescence X-ray spectrometer.

4.3.11.2 Procedure. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight of the total nonvolatile. The settings for a wavelength dispersive fluorescence spectrometer shall be as follows: (1)

Element	Analytical Line	Angle	Crystal	Detection	Collimeter	X-ray tube (MO)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd I)		33.00	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd II)		35.50	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60Kv 45Ma

Pulse height selection shall be used in all measurements and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background, and the Molybdenum Compton scattered background from the X-ray tube.

4.3.11.3 Calculation.

$$R = \frac{I_{Pb} - \frac{I_{Pb\ (Background\ I)} + I_{Pb\ (Background\ II)}}{2}}{I_{Mo}}$$

where I equals gross intensity. These results shall be compared to those obtained with a 0.06 percent lead standard made up from the same type of paint sample and evaluated for compliance with table I.

(1) Energy dispersive fluorescence spectrometers shall be set up according to the manufacturer's manual.

Paragraph 5.1, line 2. Change "TT-P-143" to "PPP-P-1892".

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